Statistics of Financial Markets

Seminar 2: Stable Distributions

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Learning Objectives

By the end of this seminar you will be able to:

- Verify the stability property for Gaussian and Cauchy
- Compute the characteristic function of a stable distribution
- Analyse moments and tail behaviour
- Estimate parameters of a stable distribution (McCulloch, MLE)
- Compare VaR: Gaussian vs Stable vs Empirical
- Implement stable fitting in Python

I. Manual exercises (25 min)
↓
II. Estimation (20 min)
↓
III. Tails & risk (15 min)
↓
IV. Python (30 min)

Exercise 1: Verifying Stability — The Gaussian Case

Problem

Let X_1, X_2 be i.i.d. $\sim \mathcal{N}(0,1)$. Show that $aX_1 + bX_2 \stackrel{d}{=} cX + d$ for $a = 3, b = 4, c = 5$.

Solution

If $X_1 \sim \mathcal{N}(0,1)$, then $aX_1 \sim \mathcal{N}(0, a^2)$ and $bX_2 \sim \mathcal{N}(0, b^2)$.
Sum: $aX_1 + bX_2 \sim \mathcal{N}(0, a^2 + b^2) = \mathcal{N}(0, 9 + 16) = \mathcal{N}(0, 25)$.
Hence $c = \sqrt{a^2 + b^2} = \sqrt{25} = 5$ and $d = 0$.

Conclusion: The normal distribution is stable with $\alpha = 2$. The stability property reduces to the variance addition rule.

Exercise 1 cont.: Cauchy Case + Comparison Table

Cauchy case ($\alpha = 1$)

Let $X_1, X_2 \sim \text{Cauchy}(0, 1)$ i.i.d.
$aX_1 + bX_2 \sim \text{Cauchy}(0, a + b)$
For $a = 3, b = 4$: $c = a + b = 7$, $d = 0$.

Note: For Cauchy, scales add *linearly*, not quadratically!

General rule

For $S(\alpha, \beta, \gamma, \delta)$:

$$c^\alpha = a^\alpha + b^\alpha$$

	α	Rule	c
Gaussian	2	$c^2 = a^2 + b^2$	5
Cauchy	1	$c = a + b$	7
$S(1.5)$	1.5	$c^{1.5} = a^{1.5} + b^{1.5}$	6.19

Exercise 2: The Characteristic Function

Problem

Compute $\varphi_X(t)$ for $X \sim S(1.5, 0.3, 1, 0)$ at $t = 1$.

Solution

The characteristic function of a stable distribution:

$$\log \varphi_X(t) = -|t|^\alpha \left[1 - i\beta \operatorname{sgn}(t) \tan \frac{\pi\alpha}{2}\right] + i\delta t$$

For $\alpha = 1.5$, $\beta = 0.3$, $\gamma = 1$, $\delta = 0$, $t = 1$:

$$\log \varphi_X(1) = -|1|^{1.5} \left[1 - i \cdot 0.3 \cdot 1 \cdot \tan \frac{3\pi}{4}\right]$$

$$= -1 \cdot [1 - i \cdot 0.3 \cdot (-1)] = -1 - 0.3i$$

Therefore:

$$\varphi_X(1) = e^{-1-0.3i} = e^{-1}(\cos 0.3 - i \sin 0.3) \approx 0.351 - 0.109i$$

$$|\varphi_X(1)| = e^{-1} \approx 0.368$$

Exercise 3: Moments and Tails

Problem

A financial analyst models daily returns with $S(1.7, -0.2, 0.01, 0.0005)$. Which moments exist? What are the risk management implications?

Solution

Moment existence:

- $\mathbb{E}[|X|^p] < \infty$ if and only if $p < \alpha$
- $\alpha = 1.7 \Rightarrow$
  - Mean ($p = 1$): exists ($1 < 1.7$)
  - Variance ($p = 2$): does not exist ($2 > 1.7$)
  - Kurtosis ($p = 4$): does not exist

Practical implications:

- $\beta = -0.2$: negative skewness (downside risk)
- Gaussian VaR **underestimates** true risk
- Probability of a -5σ event:
  - Normal: $\approx 3 \times 10^{-7}$ (1 in 3.5M days)
  - Stable(1.7): ≈ 0.003 (1 in 333 days)
- Sharpe ratio is not rigorously defined!

McCulloch's Method — Concept

Key idea

Estimate parameters $(\alpha, \beta, \gamma, \delta)$ from **sample quantiles**.

Intermediate statistics:

$$v_\alpha = \frac{q_{0.95} - q_{0.05}}{q_{0.75} - q_{0.25}}$$

$$v_\beta = \frac{q_{0.95} + q_{0.05} - 2q_{0.50}}{q_{0.95} - q_{0.05}}$$

- v_α measures *tail thickness*
- v_β measures *asymmetry*

Reference table (selection):

v_α	α	Interpretation
2.44	2.0	Gaussian
2.62	1.9	Near Gaussian
3.04	1.7	Moderate tails
4.21	1.5	Heavy tails
6.31	1.3	Very heavy tails

McCulloch — Worked Example

Numerical data

From a sample of 2,000 observations: $q_{0.05} = -3.42$, $q_{0.25} = -0.78$, $q_{0.50} = 0.05$, $q_{0.75} = 0.82$, $q_{0.95} = 3.61$.

Step-by-step solution

Step 1: Compute v_α

$$v_\alpha = \frac{q_{0.95} - q_{0.05}}{q_{0.75} - q_{0.25}} = \frac{3.61 - (-3.42)}{0.82 - (-0.78)} = \frac{7.03}{1.60} = 4.39$$

Step 2: Estimate α From the table: $v_\alpha = 4.39$ corresponds to $\hat{\alpha} \approx 1.48$.

Step 3: Compute v_β

$$v_\beta = \frac{q_{0.95} + q_{0.05} - 2q_{0.50}}{q_{0.95} - q_{0.05}} = \frac{3.61 + (-3.42) - 2 \cdot 0.05}{7.03} = \frac{0.09}{7.03} \approx 0.013$$

Step 4: Estimate β $v_\beta \approx 0.013 \approx 0 \Rightarrow \hat{\beta} \approx 0$ (nearly symmetric distribution).

Maximum Likelihood Estimation (MLE)

Procedure

$$\hat{\theta}_{MLE} = \arg\max_{\theta} \sum_{i=1}^{n} \log f(x_i; \alpha, \beta, \gamma, \delta)$$

Challenges:

- PDF has no closed form (in general)
- Requires numerical evaluation via FFT
- Likelihood function is not convex
- Slow convergence for small α

Advantages:

- Asymptotically most efficient estimator
- Standard confidence intervals
- Modern implementation: `scipy.stats.levy_stable.fit()`

Python one-liner:

```
from scipy.stats import levy_stable
a, b, loc, sc = levy_stable.fit(data)
```

Koutrouvelis Regression

Two-stage method

Uses the empirical characteristic function (ECF).

Stage 1: Estimate α and γ via linear regression:

$$\log\left(-\log|\hat{\varphi}_n(t)|^2\right) = \log(2\gamma^\alpha) + \alpha\log|t|$$

Regression $Y = a + bX$ with $Y_k = \log(-\log|\hat{\varphi}_n(t_k)|^2)$ and $X_k = \log|t_k|$.

Stage 2: Estimate β and δ via regression on the imaginary part:

$$\arg[\hat{\varphi}_n(t)] = \delta t - \beta\gamma^\alpha \tan\frac{\pi\alpha}{2}\,\mathrm{sgn}(t)|t|^\alpha$$

Advantage

Numerically robust, does not require PDF evaluation, fast convergence.

Heavy Tails — Gaussian vs Stable

SFM_ch2_stable_fit

Comparison table: Probability of extreme events

Event	Gaussian	Stable(α=1.7)	Ratio
$\lvert X\rvert > 3\sigma$	2.7×10^{-3}	1.9×10^{-2}	$\approx 7\times$
$\lvert X\rvert > 5\sigma$	5.7×10^{-7}	3.0×10^{-3}	$\approx 5{,}000\times$
$\lvert X\rvert > 10\sigma$	1.5×10^{-23}	5.2×10^{-4}	$\approx 10^{19}\times$

Practical implication

Gaussian models dramatically underestimate the risk of extreme events. A "10σ" event that the Gaussian considers virtually impossible occurs roughly once every $\sim$2,000 days ($\sim$8 years) under the stable model.

Exercise 4: Gaussian vs Stable VaR

Problem

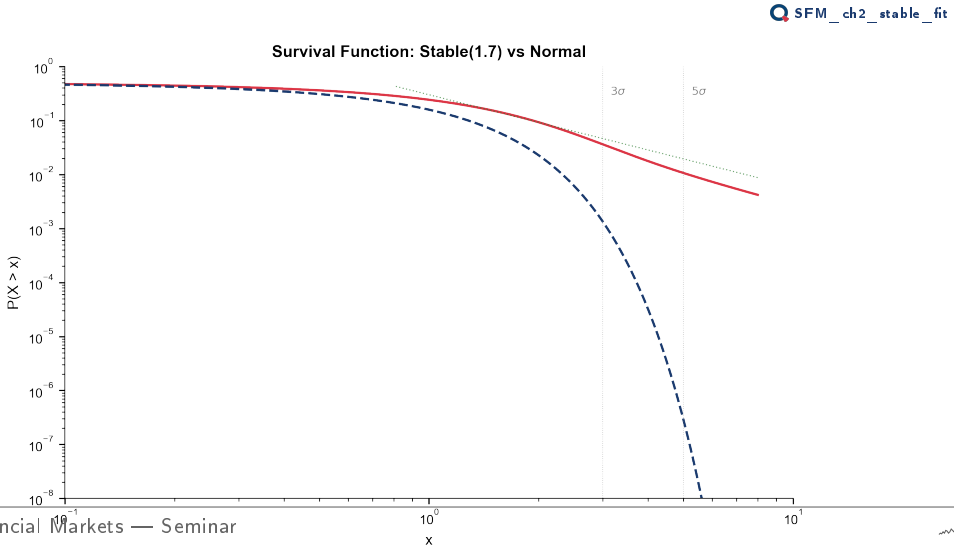
Portfolio of 1,000,000 EUR. Daily returns modelled with $S(1.7, -0.2, 0.01, 0.0005)$. Compare VaR at 1%, 5% and 0.1%.

Solution

VaR level	Gaussian	Stable	Difference
5%	−16,449 EUR	−21,340 EUR	+30%
1%	−22,763 EUR	−39,870 EUR	+75%
0.1%	−30,091 EUR	−87,420 EUR	+190%

Conclusion: The deeper into the tail, the larger the gap between Gaussian and Stable VaR. At 0.1%, the stable model estimates losses nearly **3 times larger**.

Visual Comparison: Survival Function



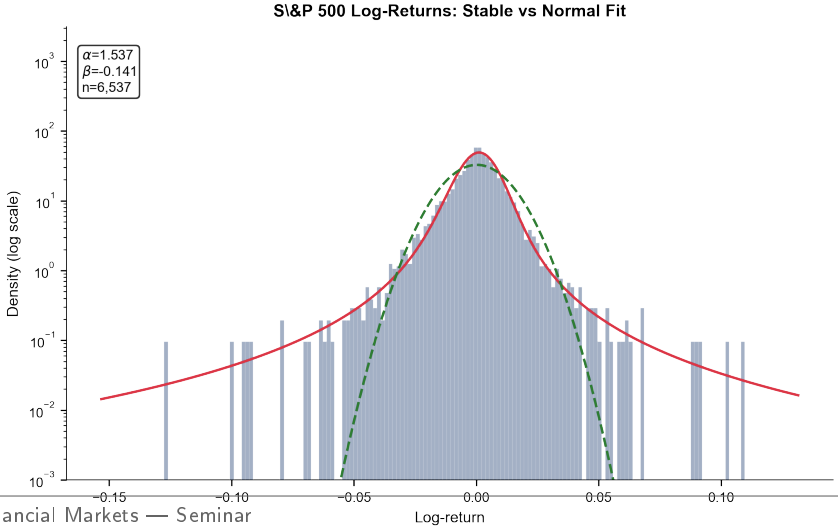
Simulating Stable Distributions

```
import numpy as np
from scipy.stats import levy_stable
import matplotlib.pyplot as plt

# Simulate stable distributions for different alpha
alphas = [0.5, 1.0, 1.5, 2.0]
x = np.linspace(-8, 8, 1000)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
colors = ['#DC3545', '#FF8C00', '#2E7D32', '#1A3A6E']
labels = [r'$\alpha=0.5$ (Levy)', r'$\alpha=1.0$ (Cauchy)',
          r'$\alpha=1.5$', r'$\alpha=2.0$ (Gaussian)']

for a, c, lab in zip(alphas, colors, labels):
    pdf = levy_stable.pdf(x, alpha=a, beta=0)
    ax1.plot(x, pdf, color=c, label=lab)       # Linear scale
    ax2.semilogy(x, pdf, color=c, label=lab)    # Log scale

ax1.set_title('PDFs (linear)')
ax2.set_title('PDFs (log scale) - tail behavior')
for ax in [ax1, ax2]:
    ax.legend(loc='upper center', bbox_to_anchor=(0.5, -0.12),
              ncol=2, frameon=False)
plt.tight_layout()
```

Fitting Stable to S&P 500

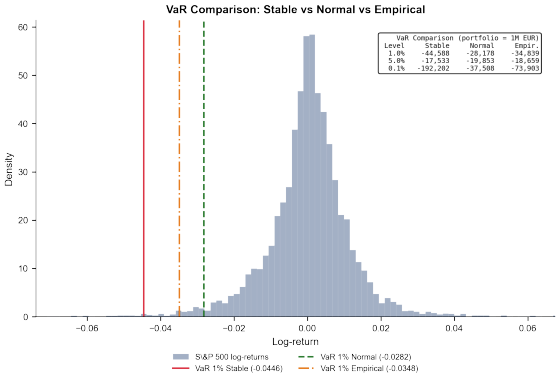
```
import yfinance as yf
from scipy.stats import levy_stable, norm

# Download S&P 500 data
sp500 = yf.download('^GSPC', start='2000-01-01', end='2025-12-31')
returns = np.log(sp500['Close'] / sp500['Close'].shift(1)).dropna()

# Fit stable distribution
alpha, beta, loc, scale = levy_stable.fit(returns.values)
print(f"Stable fit: alpha={alpha:.3f}, beta={beta:.3f}")

# Fit normal
mu, sigma = norm.fit(returns.values)
print(f"Normal fit: mu={mu:.6f}, sigma={sigma:.6f}")

# Compare on log-scale
x = np.linspace(returns.min()*1.2, returns.max()*1.2, 1000)
plt.hist(returns, bins=150, density=True, alpha=0.4, label='Data')
plt.semilogy(x, levy_stable.pdf(x, alpha, beta, loc, scale),
             label=f'Stable (alpha={alpha:.2f})')
plt.semilogy(x, norm.pdf(x, mu, sigma), '--', label='Normal')
plt.legend(loc='upper center', bbox_to_anchor=(0.5, -0.1),
           ncol=3, frameon=False)
```

Overlay PDF: Histogram + Stable + Normal (log-scale)



VaR Comparison + Backtest

```python
# VaR comparison
levels = [0.01, 0.05, 0.001]
for cl in levels:
    var_s = levy_stable.ppf(
        cl, alpha, beta, loc, scale)
    var_n = norm.ppf(cl, mu, sigma)
    var_e = np.percentile(
        returns, cl*100)
    print(f"VaR {cl*100:.1f}%:")
    print(f"  Stable:    {var_s:.5f}")
    print(f"  Normal:    {var_n:.5f}")
    print(f"  Empirical: {var_e:.5f}")
```



AI Exercise: Prompt Engineering

Exercise

Use ChatGPT or Claude to generate Python code that:

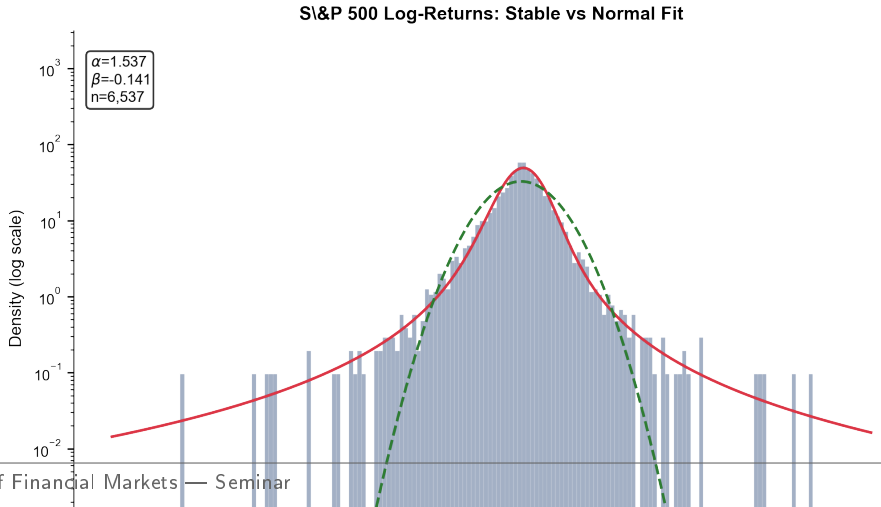
1. Downloads BTC-USD data for the last 5 years
2. Fits a stable distribution to the daily log-returns
3. Compares stable vs normal PDF graphically (log-scale)
4. Computes VaR at 1% and 5%
5. Performs a backtest: how many days exceeded the estimated VaR?

Suggested prompt

"Write Python code using scipy.stats.levy_stable to fit a stable distribution to BTC-USD daily log-returns from yfinance. Plot histogram with stable and normal overlay on log-scale. Calculate VaR at 1% and 5% for both distributions and backtest against actual returns."

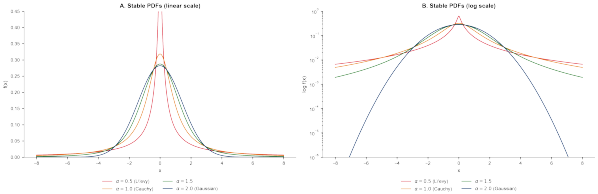
Histogram + Stable vs Normal Fit

SFM_ch2_stable_fit



PDF Comparison on Logarithmic Scale

SFM_ch2_stable_fit



Observations:

- Panel B (log scale) highlights tail differences
- $\alpha = 0.5$ (Lévy) has the heaviest tails
- At $\alpha = 2.0$ we recover the normal distribution exactly
- Financial returns typically have $\alpha \in [1.5, 1.9]$

VaR: Comparison Table

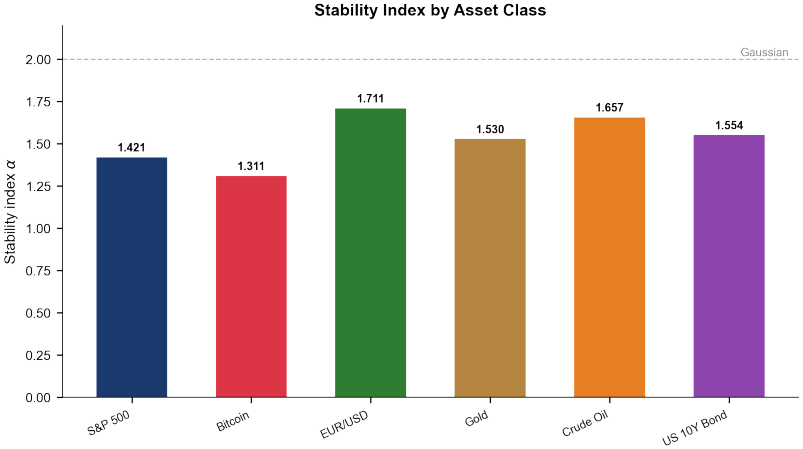
VaR on S&P 500 (2000–2025): Gaussian vs Stable vs Empirical

VaR level	Gaussian	Stable	Empirical
5%	−1.91%	−2.14%	−2.08%
1%	−2.71%	−4.12%	−3.56%
0.1%	−3.59%	−8.97%	−6.88%

- At the 5% level, differences are small
- At 1%, the stable model is ∼50% more conservative than Gaussian
- At 0.1%, the stable model estimates losses **2.5**× larger
- Empirical VaR falls **between** the two parametric models

Cross-Asset Comparison: Stability Index α

SFM_ch2_stable_fit



- Bitcoin has the lowest $\alpha \Rightarrow$ heaviest tails
- Bonds have the highest α (closest to Gaussian)

Homework: Stable Distributions on BVB

Requirements

1. Choose a stock listed on the Bucharest Stock Exchange (e.g. TLV, SNP, BRD, FP)
2. Download daily data for the last 5 years
3. Fit a stable distribution to the log-returns
4. Compare stable vs normal PDF graphically (log scale)
5. Compute VaR at 1% and 5% (stable vs normal vs empirical)
6. Interpret the results: is the Romanian market "more Gaussian" than S&P 500?

Hints

- Use `yfinance` with the BVB ticker (e.g. `TLV.RO`)
- `levy_stable.fit()` can be slow — use `floc=np.mean(data)`
- Compare the estimated α with that of S&P 500

Competence Checklist

Theory

- ☐ Stability definition via sums
- ☐ Parameters: α, β, γ, δ
- ☐ Special cases: Gauss, Cauchy, Lévy
- ☐ Characteristic function
- ☐ Moment existence ($p < \alpha$)
- ☐ Tail behaviour (power law)

Practice

- ☐ McCulloch estimation (quantiles)
- ☐ MLE with `levy_stable.fit()`
- ☐ Koutrouvelis regression (ECF)
- ☐ VaR: stable vs normal vs empirical
- ☐ Simulation and visualisation in Python
- ☐ Interpreting α on real data

Seminar Planning

Week	Topic	Associated lecture	Main content
1	Returns	Ch. 1: Data sources	Return types, stationarity
2	**Stable distributions**	**Ch. 2: Distributions**	**Stable fit, VaR, heavy tails**
3	EMH & Random Walk	Ch. 3–4	EMH tests, variance ratio
4	Stylized facts	Ch. 5–6	ACF, volatility clustering

Quiz: 5 Questions

1. What are the bounds of the stability parameter α?
   (a) $0 < \alpha \leq 1$
   (b) $0 < \alpha \leq 2$
   (c) $-2 \leq \alpha \leq 2$
   (d) $\alpha > 0$

2. For a distribution $S(1.6, 0, 1, 0)$, which moments **exist**?
   (a) Mean and variance
   (b) Only the mean
   (c) All moments
   (d) No moments

3. Which distribution is obtained at $\alpha = 1, \beta = 0$?
   (a) Gaussian
   (b) Cauchy
   (c) Lévy
   (d) Exponential

Quiz: 5 Questions (cont.) + Answers

4. In McCulloch's method, $v_\alpha = 2.44$ corresponds to:
   (a) Cauchy
   (b) Gaussian
   (c) Lévy
   (d) Stable($\alpha = 1.5$)

5. Why is stable VaR $>$ Gaussian VaR at 0.1%?
   (a) The stable distribution has heavier tails
   (b) The stable distribution is more symmetric
   (c) The Gaussian has infinite variance
   (d) The parameter β is negative

Answers

1. **(b)** $0 < \alpha \leq 2$
2. **(b)** Only the mean (since $p < \alpha = 1.6$, only $p < 1.6$)
3. **(b)** Cauchy
4. **(b)** Gaussian ($\alpha = 2.0$)
5. **(a)** Heavier tails of the stable distribution

References

[1] J.P. Nolan, *Stable Distributions: Models for Heavy Tailed Data*, Birkhäuser, 2020.

[2] B. Mandelbrot, "The variation of certain speculative prices," *Journal of Business*, 36(4), 1963.

[3] S.T. Rachev, S. Mittnik, *Stable Paretian Models in Finance*, Wiley, 2000.

[4] J. Franke, W.K. Härdle, C.M. Hafner, *Statistics of Financial Markets*, 4th ed., Springer, 2019.

[5] J.H. McCulloch, "Simple consistent estimators of stable distribution parameters," *Communications in Statistics — Simulation and Computation*, 15(4), 1986.

Online resources:

- `https://quantlet.com` — Quantlet repository
- `https://danpele.github.io/SFM/` — Course materials

Thank you!

Questions?

Course materials available at: https://danpele.github.io/SFM/

Quantlet Quantinar